

ELEC3705 Fundamentals of Quantum Engineering

Laboratory 3 : Time evolution of an electron spin

Abstract

In this lab we will simulate an electron spin and its evolution under a time-independent magnetic field. We will observe the effect known as Larmor precession.

Introduction

An electron is a spin-half particle, which will be our ongoing example of a simple 2 level quantum system. Equation 1 defines the spin matrix vector for an electron. Be aware that each vector component is a matrix. When a spin is placed in a magnetic field, it introduces an energy splitting, or a bias between two states. In the past, we assumed these states were $|\uparrow\rangle$ and $|\downarrow\rangle$, referenced to the $\hat{z}$ -axis. Without an applied magnetic field, these states would be indistinguishable, or degenerate. To make this clear, refer back to the Stern-Gerlach experiment when the magnetic field was turned off. In this arrangement there was only a single line where the particles were detected. Both spin states were ‘measured’ with the same outcome, therefore they were indistinguishable. As will be revealed in the next section, this degeneracy is due to the two measurable states having the same energy.

In the previous lab, we briefly looked into how measurements along different directions changed our quantum state. Each measurement had a discrete and instantaneous outcome. This lab will explore how dynamical, time-continuous effects can arise and how we can describe this mathematically.

The Hamiltonian Operator

$$\begin{aligned}\mathbf{S} &= \frac{\hbar}{2}\boldsymbol{\sigma} \\ &= \frac{\hbar}{2}\begin{pmatrix} \sigma_x & \sigma_y & \sigma_z \end{pmatrix}\end{aligned}\tag{1}$$

Recall that a system Hamiltonian is a description of the total energy of the system, which is typically defined through so-called interaction terms. An electron has an intrinsic magnetic moment (spin) which interacts with an external magnetic field. The simplest interaction is known as the linear Zeeman interaction, which we use below. The Zeeman effect for a single

spin can be defined as in equation 2.

$$\begin{aligned}\mathcal{H}(E) &= 2\pi \gamma_e \mathbf{B} \cdot \mathbf{S} \\ &= 2\pi \gamma_e \frac{\hbar}{2} (B_x \sigma_x + B_y \sigma_y + B_z \sigma_z) [\text{J}]\end{aligned}\tag{2}$$

where γ_e is the gyromagnetic ratio of an electron and is approximately 28 GHz T^{-1} .

We can scale our Hamiltonian to express it in terms of frequency, an engineer's bread and butter.

$$\begin{aligned}\mathcal{H}(\omega) &= \frac{2\pi \gamma_e}{\hbar} \mathbf{B} \cdot \mathbf{S} \\ &= \pi \gamma_e (B_x \sigma_x + B_y \sigma_y + B_z \sigma_z) [\text{rad s}^{-1}]\end{aligned}\tag{3}$$

This will prove very useful when we observe the time evolution of a state, as we can immediately understand the system in terms of its frequencies.

Question 1

Consider a single electron under the influence of a magnetic field, $B_z = 1 \text{ T}$ and $B_x = B_y = 0 \text{ T}$. Calculate the Hamiltonian and its eigenvalues for this system in units of frequency.

Time Evolution

We can observe how a given state evolves in a system defined by a Hamiltonian, $\mathcal{H}$, through the use of the Schrödinger equation.

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \mathcal{H}(E) |\psi(t)\rangle\tag{4}$$

Equation 5 shows a basic formulation of time evolution which uses the inner product to measure the ‘projection’ of your state vector, $|\psi(t)\rangle$, onto the eigenvectors of $\mathcal{H}$, $|\varphi_n\rangle$. Therefore, it is quite easy to calculate each $c_n(t)$ over a time interval and hence calculate your ‘evolved’ state of $|\psi\rangle$ ¹.

$$|\psi(t)\rangle = \sum_n c_n(t) |\varphi_n\rangle\tag{5}$$

where

$$c_n(t=0) = \langle \varphi_n | \psi(t=0) \rangle\tag{6}$$

$$c_n(t) = c_n(t_0) e^{-i\omega_n(t-t_0)}\tag{7}$$

¹This is simply another way of viewing the time evolution operator $U(t, t' = 0) = \exp(-iH(\omega)t)$

Question 2

Given the Hamiltonian

$$\mathcal{H} = 2\pi \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} [\text{rad s}^{-1}]$$

find its eigenvectors and then calculate the coefficients of time evolution at the following times, and with the initial state $|\psi(t=0)\rangle = |\uparrow\rangle$.

$$t = [0, 0.1 \text{ s}, 0.2 \text{ s}, \dots, 1 \text{ s}]$$

Once you have this, calculate the time evolved state's overlap with the initial state, i.e. $|\langle\psi(t)|\psi(t=0)\rangle|^2$. Give your interpretation of the results.

Question 3

Using the Hamiltonian given in question 1, determine an expression for $|\psi(t)\rangle$, given

$$|\psi(t=0)\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$$

Question 4

Finally, derive an expression for the time evolving expectation values $\langle S_x \rangle_{\psi(t)}$, $\langle S_y \rangle_{\psi(t)}$ and $\langle S_z \rangle_{\psi(t)}$

Task

Using the Hamiltonian described in equation 3, calculate the energy of an electron spin in Earth's magnetic field. Assume that $|\mathbf{B}| = 57 \mu\text{T}$, and has an angle of inclination of 64° to the surface of Earth. Calculate the Larmor precession frequency of the electron spin, and plot a single period of evolution for the given starting states.

a) $|\psi(t=0)\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$

b) $|\psi(t=0)\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle)$

c) $|\psi(t=0)\rangle = |\uparrow\rangle$

Extended Task

Repeat the simulation as in the previous section, but plot the results using a Husimi Q representation plotted on a Bloch sphere. This now shows the spin orientation as a colour-map of the spin projection along all directions in 3D space, not just the point with maximum spin projection we've considered before. This is not examinable, however it offers a stronger intuition of the concept of a quantum state.

Question 5

Why does the state vector appear as a smudge on the Bloch sphere? (i.e. why is there a non-zero probability of the state vector being measured in different states)